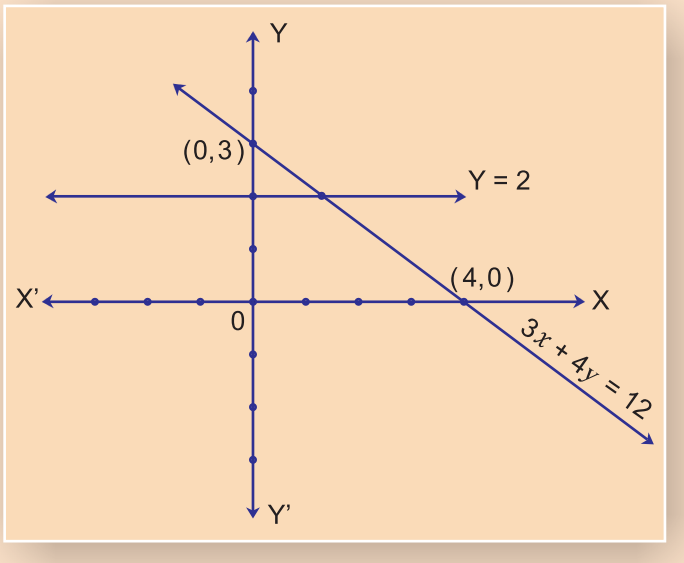


Mathematics

Part - I

STANDARD TEN





$$1 + 2 + 3 + \cdots + \cdots + 78 + 79 + 80$$

$$= (1 + 80) + (2 + 79) + \cdots + (39 + 42) + (40 + 41)$$

The Constitution of India

Chapter IV A

Fundamental Duties

ARTICLE 51A

Fundamental Duties- It shall be the duty of every citizen of India–

(a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;

(b) to cherish and follow the noble ideals which inspired our national struggle for freedom;

(c) to uphold and protect the sovereignty, unity and integrity of India;

(d) to defend the country and render national service when called upon to do so;

(e) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities, to renounce practices derogatory to the dignity of women;

(f) to value and preserve the rich heritage of our composite culture;

(g) to protect and improve the natural environment including forests, lakes, rivers and wild life and to have compassion for living creatures;

(h) to develop the scientific temper, humanism and the spirit of inquiry and reform;

(i) to safeguard public property and to abjure violence;

(j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement;

(k) who is a parent or guardian to provide opportunities for education to his child or, as the case may be, ward between the age of six and fourteen years.

The Coordination Committee formed by GR No. Abhyas - 2116/(Pra.Kra.43/16) SD - 4
Dated 25.4.2016 has given approval to prescribe this textbook in its meeting held on
29.12.2017 and it has been decided to implement it from the educational year 2018-19.

Mathematics

Part I

STANDARD TEN

Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune - 411 004

The digital textbook can be obtained through DIKSHA App on a smartphone by using the Q. R. Code given on title page of the textbook and useful audio-visual teaching-learning material of the relevant lesson will be available through the Q. R. Code given in each lesson of this textbook.

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The Constitution of India

Preamble

WE, THE PEOPLE OF INDIA, having solemnly resolved to constitute India into a SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC and to secure to all its citizens:

JUSTICE, social, economic and political;

LIBERTY of thought, expression, belief, faith and worship;

EQUALITY of status and of opportunity; and to promote among them all

FRATERNITY assuring the dignity of the individual and the unity and integrity of the Nation;

IN OUR CONSTITUENT ASSEMBLY this twenty-sixth day of November, 1949, do HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.

NATIONAL ANTHEM

Jana-gana-mana-adhināyaka jaya hē
Bhārata-bhāgya-vidhātā,

Panjāba-Sindhu-Gujarāta-Marāthā
Drāvida-Utkala-Banga

Vindhya-Himāchala-Yamunā-Gangā
uchchala-jaladhi-taranga

Tava subha nāmē jāgē, tava subha āsisa māgē,
gāhē tava jaya-gāthā,

Jana-gana-mangala-dāyaka jaya hē
Bhārata-bhāgya-vidhātā,

Jaya hē, Jaya hē, Jaya hē,
Jaya jaya jaya, jaya hē.

PLEDGE

India is my country. All Indians are my brothers and sisters.

I love my country, and I am proud of its rich and varied heritage. I shall always strive to be worthy of it.

I shall give my parents, teachers and all elders respect, and treat everyone with courtesy.

To my country and my people, I pledge my devotion. In their well-being and prosperity alone lies my happiness.

Preface

Dear Students,

Welcome to the tenth standard!

This year you will study two text books - Mathematics Part-I and Mathematics Part-II

The main areas in the book Mathematics Part-I are Algebra, Graph, Financial planning and Statistics. All of these topics were introduced in the ninth standard. This year you will study some more details of the same. The new tax system, GST is introduced in Financial planning. Wherever a new unit, formula or application is introduced, its lucid explanation is given. Each chapter contains illustrative solved examples and sets of questions for practice. In addition, some questions in practice sets are star-marked, indicating that they are challenging for talented students. After tenth standard, some students do not opt for Mathematics. This book gives them basic concepts and mathematics needed to work in other fields. The matter under the head 'For more information' is useful for those students who wish to study mathematics after tenth standard and achieve proficiency in it. So they are earnestly advised to study it. Read the book thoroughly at least once and understand the concepts.

Additional useful audio-visual material regarding each lesson will be available to you by using Q.R. code through 'App'. It will definitely be useful to you for your studies.

Much importance is given to the tenth standard examination. You are advised not to take the stress and study to the best of your ability to achieve expected success. Best wishes for it !



(Dr. Sunil Magar)
Director
Maharashtra State Bureau of Textbook
Production and Curriculum Research, Pune.

Pune
Date : 18 March 2018, Gudhipadva
Indian Solar Year : 27 Falgun 1939

It is expected that students will develop the following competencies after studying Mathematics– Part I syllabus in standard X

Area	Topic	Competency Statements
1. Knowledge of numbers	1.1 Arithmetic Progression	The students will be able to- • solve examples using Arithmetic Progression. • plan steps to achieve a goal in future.
2. Algebra	2.1 Quadratic Equations 2.2 Linear equations in two variables	• solve day to day problems which can be expressed in the form of quadratic equations. • decide the number of variables required to find solutions of word problems. • convert a word problem into an equation in two variables and find its solution.
3. Commercial Mathematics	3.1 Financial planning	• understand the concepts of savings and investments. • get familiar with financial transactions in business, profession etc.
4. Statistics and Probability	4.1 Probability 4.2 Graph and measures of central tendencies	• use the concept of probaility in games, voting etc. • present the collected data in the form of graphs or pictures deciding the suitable form of presentation. • find the mean, median and mode of a provided classified data.

Instructions for Teachers

Read the book in detail and understand the content thoroughly.Take the help of activities to explain different topics, to verify the formulae etc.

Practicals is also a means of evaluation. Activities given can also be used for this purpose. Encourage the students to think independently.Compliment a student if he solves an example by a different and logically correct method.

List of some practicals (Specimen)

1. On a graph paper, draw a line parallel to the X-axis or Y-axis. Write coordinates of any four points on the line. Write how the equation of the line can be obtained from the coordinates.

   [Instead of parallel lines, lines passing through the origin or intersecting the X or Y- axis can also be considered]

2. Bear a two- digit number in mind. Without disclosing it, cunstruct a puzzle. Create two algebraic relations between the two digits of the number and solve the puzzle.

   [The above practical can be extended to a three-digit number also.]

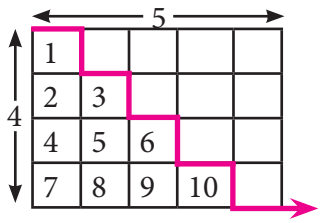
3. Read the information about contents on a food packet. Show the information by a pie diagram. For example, see the chart of contents like carbohydrates, proteins, vitamines etc. per given weight on a buiscuit packet. Show the proportion of the contents by a pie diagram. The contents can be divided into four classes as carbohydrates, proteins, fats and others.

4. Prepare a frequncy distribution table given by the teacher in Excel sheet on a computer. From the table draw a frequency polygon and a histogram in Excel.

5. Roll a die ten times and record the outcomes in the form of a table.

6. Observe the tax invoice given by your teacher. Record all of its contents. Recalculate the taxes and verify their correctness.

7. Calculate the sum of first n natural numbers given by your teacher through the following activity. For example to find the sum of first four natural numbers, take a square-grid piece of paper of 4×5 squares. Then cut it as shown in the figure. Hence verify the formula $S_n = \frac{n(n+1)}{2}$ (Here $n = 4$)



$S_n = \frac{n(n+1)}{2}$ $\therefore S_4 = \frac{4\ (4+1)}{2} = \frac{4\times5}{2} = \frac{20}{2} = 10$

[Note: Here $a = 1$ and $d = 1$. The activity can be done taking different values of a and d. Similarly, you can find the sum of even or odd numbers, cubes of natural numbers etc.]

8. Write $\alpha = 6$ on one side of a card sheet and $\alpha = -6$ on its backside. Similarly, write $\beta = -3$ on one side of another card sheet and $\beta = 7$ on its backside. From these values, form different values of $(\alpha + \beta)$ and $(\alpha\beta)$; using these values form quadratic equations.

INDEX

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1 Linear Equations in Two Variables



Let's study.

- Methods of solving linear equations in two variables – graphical method, Cramer's method
- Equations that can be transformed in linear equation in two variables
- Application of simultaneous equations



Let's recall.

Linear equation in two variables

An equation which contains two variables and the degree of each term containing variable is one, is called a linear equation in two variables.

$ax + by + c = 0$ is the general form of a linear equation in two variables; a, b, c are real numbers and a, b are not equal to zero at the same time.

Ex. $3x - 4y + 12 = 0$ is the general form of equation $3x = 4y - 12$

Activity : Complete the following table

No.	Equation	Is the equation a linear equation in 2 variables ?
1	$4m + 3n = 12$	Yes
2	$3x^2 - 7y = 13$	
3	$\sqrt{2}\,x - \sqrt{5}\,y = 16$	
4	$0x + 6y - 3 = 0$	
5	$0.3x + 0y - 36 = 0$	
6	$\frac{4}{x} + \frac{5}{y} = 4$	
7	$4xy - 5y - 8 = 0$	

Simultaneous linear equations

When we think about two linear equations in two variables at the same time, they are called simultaneous equations.

Last year we learnt to solve simultaneous equations by eliminating one variable. Let us revise it.

Ex. (1) Solve the following simultaneous equations.

(1) $5x - 3y = 8;\ 3x + y = 2$

Solution :

Method I : $5x - 3y = 8 \ldots$ (I)

$3x + y = 2 \ldots$ (II)

Multiplying both sides of equation (II) by 3.

$9x + 3y = 6 \ldots$ (III)

$5x - 3y = 8 \ldots$ (I)

Now let us add equations (I) and (III)

$$\begin{array}{r} 5x - 3y = 8 \\ +\ 9x + 3y = 6 \\ \hline 14x = 14 \end{array}$$

$\therefore\ x = 1$

substituting $x = 1$ in equation (II)

$3x + y = 2$

$\therefore\ 3 \times 1 + y = 2$

$\therefore\ 3 + y = 2$

$\therefore\ y = -1$

solution is $x = 1,\ y = -1$; it is also written as $(x, y) = (1, -1)$

Method **(II)**

$5x - 3y = 8 \ldots$ (I)

$3x + y = 2 \ldots$ (II)

Let us write value of y in terms of x from equation (II) as

$y = 2 - 3x \ldots$ (III)

Substituting this value of y in equation (I).

$5x - 3y = 8$

$\therefore\ 5x - 3(2 - 3x) = 8$

$\therefore\ 5x - 6 + 9x = 8$

$\therefore\ 14x - 6 = 8$

$\therefore\ 14x = 8 + 6$

$\therefore\ 14x = 14$

$\therefore\ x = 1$

Substituting $x = 1$ in equation (III).

$y = 2 - 3x$

$\therefore\ y = 2 - 3 \times 1$

$\therefore\ y = 2 - 3$

$\therefore\ y = -1$

$x = 1,\ y = -1$ is the solution.

Ex. (2) Solve : $3x + 2y = 29$; $5x - y = 18$

Solution : $3x + 2y = 29$. . . (I) and $5x - y = 18$ . . . (II)

Let's solve the equations by eliminating 'y'. Fill suitably the boxes below. Multiplying equation (II) by 2.

$\therefore \quad 5x \times \square - y \times \square = 18 \times \square$

$\therefore \quad 10x - 2y = \square$ . . . (III)

Let's add equations (I) and (III)

$$\begin{array}{r} 3x + 2y = 29 \\ + \square - \square = \square \\ \hline \square = \square \end{array}$$

$\therefore \quad x = \square$

Substituting $x = 5$ in equation (I)

$3x + 2y = 29$

$\therefore \quad 3 \times \square + 2y = 29$

$\therefore \quad \square + 2y = 29$

$\therefore \quad 2y = 29 - \square$

$\therefore \quad 2y = \square \qquad \therefore \quad y = \square$

$(x, y) = (\square, \square)$ is the solution.

Ex. (3) Solve : $15x + 17y = 21$; $17x + 15y = 11$

Solution : $15x + 17y = 21$. . . (I)

$17x + 15y = 11$ . . . (II)

In the two equations above, the coefficients of x and y are interchanged. While solving such equations we get two simple equations by adding and subtracting the given equations. After solving these equations, we can easily find the solution.

Let's add the two given equations.

$$\begin{array}{r} 15x + 17y = 21 \\ + 17x + 15y = 11 \\ \hline 32x + 32y = 32 \end{array}$$

Dividing both sides of the equation by 32.

$x + y = 1$ . . . (III)

Now, let's subtract equation (II) from (I)

$$\begin{array}{r} 15x + 17y = 21 \\ -\ _17x \pm 15y = _11 \\ \hline -2x + 2y = 10 \end{array}$$

dividing the equation by 2.

$-x + y = 5$ . . . (IV)

Now let's add equations (III) and (V).

$$\begin{array}{r} x + y = 1 \\ +\ -x + y = 5 \\ \hline \therefore\ 2y = 6 \end{array} \qquad \therefore \quad y = 3$$

Place this value in equation (III).

$x + y = 1$

$\therefore\ x + 3 = 1$

$\therefore\ x = 1 - 3 \qquad \therefore\ x = -2$

$(x, y) = (-2, 3)$ is the solution.

Practice Set 1.1

(1) Complete the following activity to solve the simultaneous equations.

$5x + 3y = 9$ -----(I)

$2x - 3y = 12$ ----- (II)

Let's add eqations (I) and (II).

$$\begin{array}{r} 5x + 3y = 9 \\ +\ 2x - 3y = 12 \\ \hline \square\, x = \square \end{array}$$

$x = \dfrac{\square}{\square} \qquad x = \square$

Place $x = 3$ in equation (I).

$5 \times \square + 3y = 9$

$3y = 9 - \square$

$3y = \square$

$y = \dfrac{\square}{3}$

$y = \square$

$\therefore$ Solution is $(x, y) = (\square, \square)$.

2. Solve the following simultaneous equations.

(1) $3a + 5b = 26;\ a + 5b = 22$ (2) $x + 7y = 10;\ 3x - 2y = 7$

(3) $2x - 3y = 9;\ 2x + y = 13$ (4) $5m - 3n = 19;\ m - 6n = -7$

(5) $5x + 2y = -3;\ x + 5y = 4$ (6) $\frac{1}{3}x + y = \frac{10}{3};\ 2x + \frac{1}{4}y = \frac{11}{4}$

(7) $99x + 101y = 499;\ 101x + 99y = 501$

(8) $49x - 57y = 172;\ 57x - 49y = 252$



Let's recall.

Graph of a linear equation in two variables

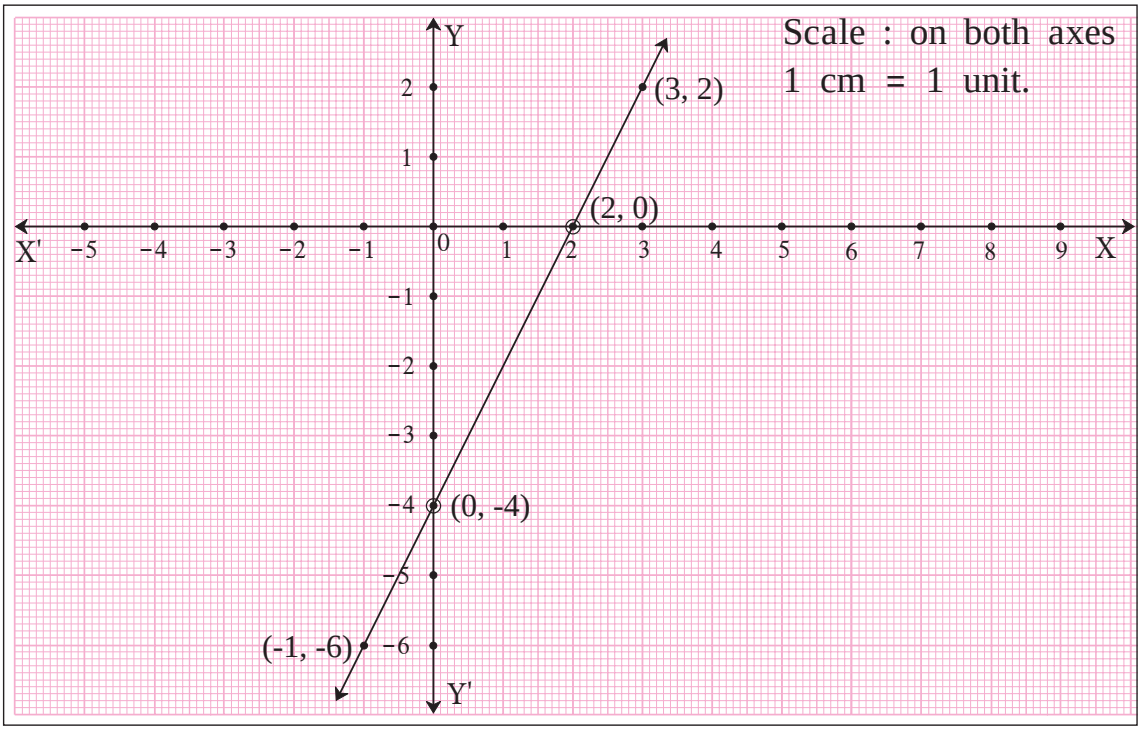
In the 9th standard we learnt that the graph of a linear equation in two variables is a straight line. The ordered pair which satisfies the equation is a solution of that equation. The ordered pair represents a point on that line.

Ex. Draw graph of $2x - y = 4$.

Solution : To draw a graph of the equation let's write 4 ordered pairs.

x	0	2	3	−1
y	−4	0	2	−6
(x, y)	$(0, -4)$	$(2, 0)$	$(3, 2)$	$(-1, -6)$

To obtain ordered pair by simple way let's take $x = 0$ and then $y = 0$.



Steps to follow for drawing a graph of linear equation in two variables.

Find at least 4 ordered pairs for given equation

↓

Draw X− axis, Y−axis on graph paper and plot the points

↓

See that all 4 points lie on one line

Two points are sufficient to represent a line, but if co−ordinates of one of the two points are wrong then you will not get a correct line.

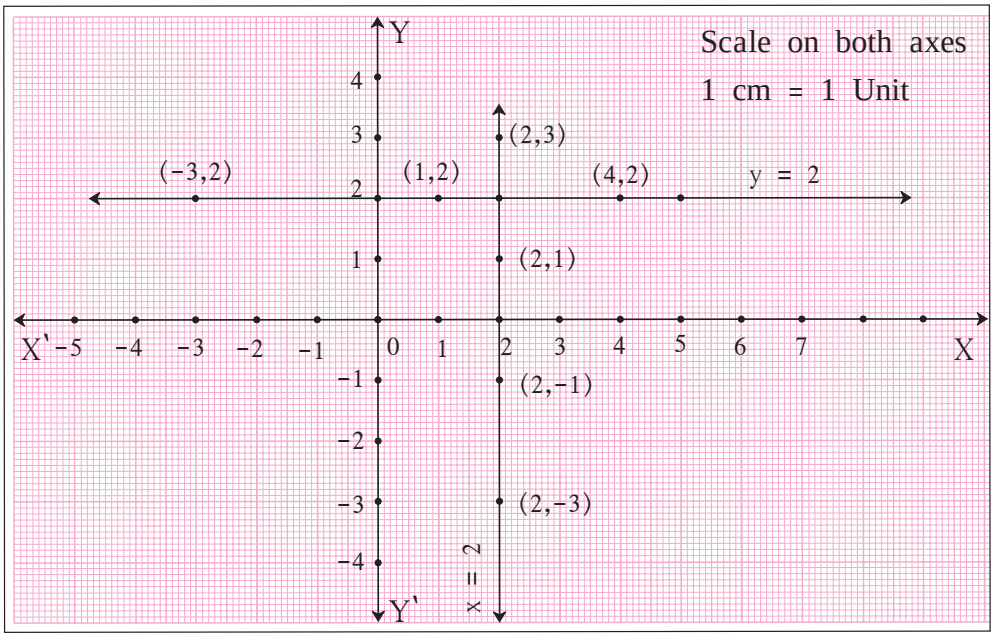
If you plot three points and if they are non collinear then it is understood that one of the points is wrongly plotted. But it is not easy to identify the incorrect point.

If we plot four points, it is almost certain that three of them will be collinear.

A linear equation $y = 2$ is also written as $0x + y = 2$. The graph of this line is parallel to X- axis; as for any value of x, y is always 2.

x	1	4	−3
y	2	2	2
(x, y)	(1, 2)	(4, 2)	(−3, 2)

Similarly equation $x = 2$ is written as $x + 0y = 2$ and its graph is parallel to Y-axis.



Let's learn.

Graphical method

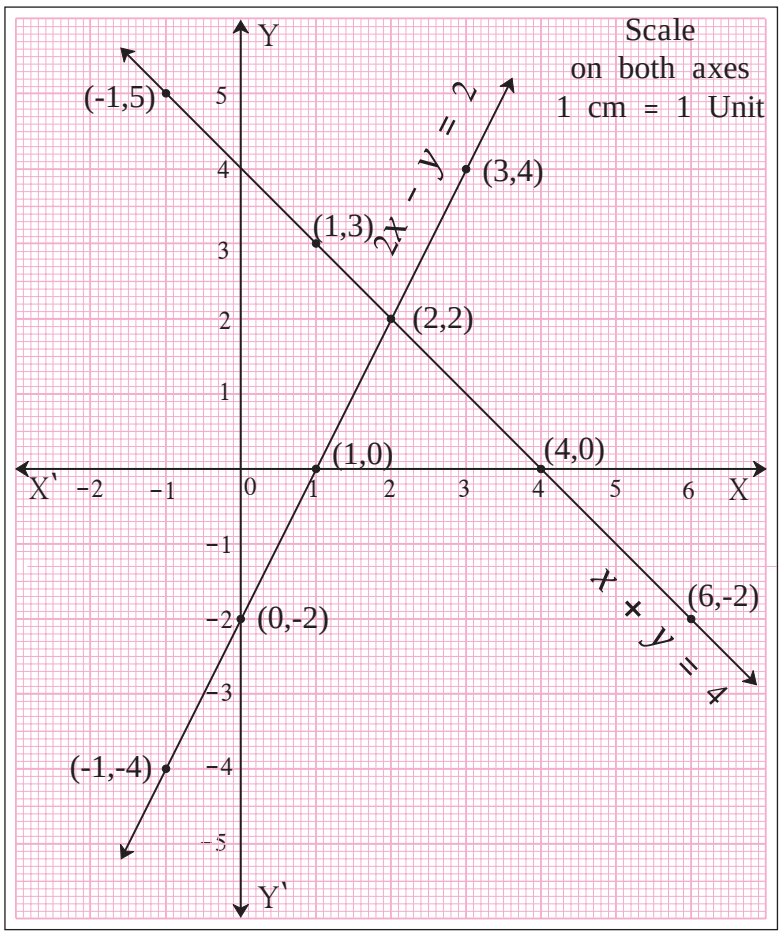
Ex. Let's draw graphs of $x + y = 4$, $2x - y = 2$ and observe them.

$x + y = 4$

x	-1	4	1	6
y	5	0	3	-2
(x, y)	$(-1, 5)$	$(4, 0)$	$(1, 3)$	$(6, -2)$

$2x - y = 2$

x	0	1	3	-1
y	-2	0	4	-4
(x, y)	$(0, -2)$	$(1, 0)$	$(3, 4)$	$(-1, -4)$



Each point on the graph satisfies the equation. The two lines intersect each other at (2, 2).

Hence ordered pair (2, 2) i.e. $x = 2$, $y = 2$ satisfies the equations $x + y = 4$ and $2x - y = 2$.

The values of variables that satisfy the given equations, give the solution of given equations.

$\therefore$ the solution of given equations $x + y = 4$, $2x - y = 2$ is $x = 2$, $y = 2$.

Let's solve these equations by method of elimination.

$x + y = 4$ . . . (I)

$2x - y = 2$ . . . (II)

Adding equations (I) and (II) we get,

$3x = 6 \quad \therefore \quad x = 2$

substituting this value in equation (I)

$x + y = 4$

$\therefore \quad 2 + y = 4$

$\therefore \quad y = 2$

Activity (I) : Solve the following simultaneous equations by graphical method.

- Complete the following tables to get ordered pairs.

$x - y = 1$

x	0		3	
y		0		-3
(x, y)				

$5x - 3y = 1$

x	2			-4
y		8	-2	
(x, y)				

- Plot the above ordered pairs on the same co-ordinate plane.
- Draw graphs of the equations. • Note the co-ordinates of the point of intersection of the two graphs. Write solution of these equations.

Activity II : Solve the above equations by method of elimination. Check your solution with the solution obtained by graphical method.



Let's think.

The following table contains the values of x and y co-ordinates for ordered pairs to draw the graph of $5x - 3y = 1$

x	0	$\frac{1}{5}$	1	-2
y	$-\frac{1}{3}$	0	$\frac{4}{3}$	$-\frac{11}{3}$
(x, y)	$\left(0, -\frac{1}{3}\right)$	$\left(\frac{1}{5}, 0\right)$	$\left(1, \frac{4}{3}\right)$	$\left(-2, -\frac{11}{3}\right)$

- Is it easy to plot these points ? • Which precaution is to be taken to find ordered pairs so that plotting of points becomes easy ?

Practice Set 1.2

1. Complete the following table to draw graph of the equations –

(I) $x + y = 3$ (II) $x - y = 4$

$x + y = 3$

x	3	☐	☐
y	☐	5	3
(x, y)	(3, 0)	☐	(0, 3)

$x - y = 4$

x	☐	-1	0
y	0	☐	-4
(x, y)	☐	☐	(0, -4)

2. Solve the following simultaneous equations graphically.

(1) $x + y = 6$; $x - y = 4$ (2) $x + y = 5$; $x - y = 3$

(3) $x + y = 0$; $2x - y = 9$ (4) $3x - y = 2$; $2x - y = 3$

(5) $3x - 4y = -7$; $5x - 2y = 0$ (6)★ $2x - 3y = 4$; $3y - x = 4$



Let's discuss.

To solve simultaneous equations $x + 2y = 4$; $3x + 6y = 12$ graphically, following are the ordered pairs.

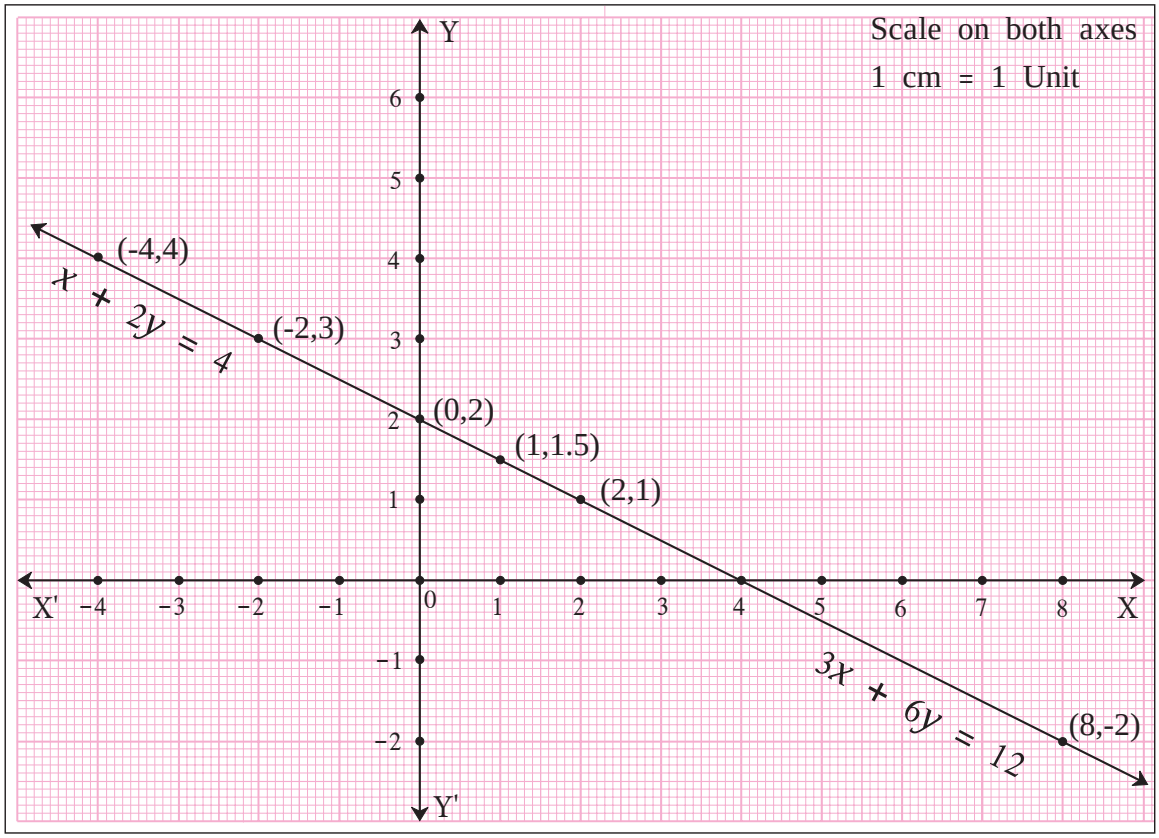
$x + 2y = 4$

x	-2	0	2
y	3	2	1
(x, y)	$(-2, 3)$	$(0, 2)$	$(2, 1)$

$3x + 6y = 12$

x	-4	1	8
y	4	1.5	-2
(x, y)	$(-4, 4)$	$(1, 1.5)$	$(8, -2)$

Plotting the above ordered pairs, graph is drawn. Observe it and find answers of the following questions.



(1) Are the graphs of both the equations different or same ?

(2) What are the solutions of the two equations $x + 2y = 4$ and $3x + 6y = 12$? How many solutions are possible ?

(3) What are the relations between coefficients of x, coefficients of y and constant terms in both the equations ?

(4) What conclusion can you draw when two equations are given but the graph is only one line ?

Now let us consider another example.

Draw graphs of $x - 2y = 4$, $2x - 4y = 12$ on the same co-ordinate plane. Observe it. Think of the realation between the coefficients of x, coefficients of y and the constant terms and draw the inference.



ICT Tools or Links.

Use Geogebra software, draw X- axis, Y-axis. Draw graphs of simultaneous equations.



Let's learn.

Determinant

$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is a determinant. (a, b), (c, d) are rows and $\begin{pmatrix} a \\ c \end{pmatrix}$, $\begin{pmatrix} b \\ d \end{pmatrix}$ are columns.

Degree of this determinant is 2, because there are 2 elements in each column and 2 elements in each row. Determinant represents a number which is $(ad\text{-}bc)$.

i.e. $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad\text{-}bc$

$ad\text{-}bc$ is the value of determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$

Determinants, usually, are represented with capital letters as A, B, C, D, ... etc.

Solved Examples

Ex. Find the values of the following determinants.

(1) A $= \begin{vmatrix} 5 & 3 \\ 7 & 9 \end{vmatrix}$ (2) N $= \begin{vmatrix} -8 & -3 \\ 2 & 4 \end{vmatrix}$ (3) B $= \begin{vmatrix} 2\sqrt{3} & 9 \\ 2 & 3\sqrt{3} \end{vmatrix}$

Solution :

(1) $A = \begin{vmatrix} 5 & 3 \\ 7 & 9 \end{vmatrix} = (5 \times 9) - (3 \times 7) = 45 - 21 = 24$

(2) $N = \begin{vmatrix} -8 & -3 \\ 2 & 4 \end{vmatrix} = [(-8) \times (4)] - [(-3) \times 2)] = -32 - (-6)$

$= -32 + 6 = -26$

(3) $B = \begin{vmatrix} 2\sqrt{3} & 9 \\ 2 & 3\sqrt{3} \end{vmatrix} = [2\sqrt{3} \times 3\sqrt{3})] - [2 \times 9)] = 18 - 18 = 0$



Let's learn.

Determinant method (Cramer's Rule)

Using determinants, simultaneous equaions can be solved easily and in less space. This method is known as determinant method. This method was first given by a Swiss mathematician Gabriel Cramer, so it is also known as Cramer's method.

To use Cramer's method, the equations are written as $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$.

$$a_1x + b_1y = c_1 \quad \ldots \quad (I)$$

$$a_2x + b_2y = c_2 \quad \ldots \quad (II)$$

Here x and y are variables, a_1, b_1, c_1 and a_2, b_2, c_2 are real numbers, $a_1b_2 - a_2b_1 \neq 0$

Now let us solve these equations.

Multiplying equation (I) by b_2.

$$a_1b_2x + b_1b_2y = c_1b_2 \quad \ldots \quad (III)$$

Multiplying equation (II) by b_1.

$$a_2b_1x + b_2b_1y = c_2b_1 \quad \ldots \quad (IV)$$

Subtracting equation (IV) from (III)

$$a_1 b_2 x + b_1 b_2 y = c_1 b_2$$
$$- \quad a_2 b_1 x - b_2 b_1 y = c_2 b_1$$

$$(a_1 b_2 - a_2 b_1)\ x = c_1 b_2 - c_2 b_1$$

$$x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1} \quad \ldots \quad (V)$$

Similarly $y = \dfrac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1} \quad \ldots \quad (VI)$

To remember and write the expressions

$c_1 b_2 - c_2 b_1$, $a_1 b_2 - a_2 b_1$, $a_1 c_2 - a_2 c_1$ we use the determinants.

Now $a_1 x + b_1 y = c_1$

and $a_2 x + b_2 y = c_2$

We can write 3 columns. $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

The values x, y in equation (V), (VI) are written using determinants as follows

$$x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1} = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}},$$

$$y = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1} = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

To remember let us denote $D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$, $D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$, $D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$

$$\therefore\ x = \frac{D_x}{D},\quad y = \frac{D_y}{D}$$

For writting D, Dx, Dy remember the order of columns $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.

From the equations,

$a_1 x + b_1 y = c_1$

and $a_2 x + b_2 y = c_2$ we get the columns $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$, $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.

- In D the column of constants $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$ is omitted.
- In D_x the column of the coefficients of x, $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ is replaced by $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.
- In D_y the column of the coefficients of y, $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ is replaced by $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.



Let's remember!

Cramer's method to solve simultaneous equations.

Write given equations in the form $ax + by = c$.

↓

Find the values of determinants D, D_x and D_y

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Using, $x = \frac{D_x}{D}$ and $y = \frac{D_y}{D}$

find values of x, y.

Gabriel Cramer

(31 July, 1704 to 4 January, 1752)

This Swiss mathematician was born in Geneva. He was very well versed in mathematics, since childhood. At the age of eighteen, he got a doctorate. He was a professor in Geneva.



Solved Example

Ex. (1) Solve the following simultaneous equations using Cramer's Rule.

$5x + 3y = -11$; $2x + 4y = -10$

Solution : Given equations

$5x + 3y = -11$

$2x + 4y = -10$

$$D = \begin{vmatrix} 5 & 3 \\ 2 & 4 \end{vmatrix} = (5 \times 4) - (2 \times 3) = 20 - 6 = 14$$

$$D_x = \begin{vmatrix} -11 & 3 \\ -10 & 4 \end{vmatrix} = (-11) \times 4 - (-10) \times 3 = -44 - (-30)$$

$$= -44 + 30 = -14$$

$$D_y = \begin{vmatrix} 5 & -11 \\ 2 & -10 \end{vmatrix} = 5 \times (-10) - 2 \times (-11) = -50 - (-22)$$

$$= -50 + 22 = -28$$

$x = \frac{D_x}{D} = \frac{-14}{14} = -1$ | and $y = \frac{D_y}{D} = \frac{-28}{14} = -2$

$\therefore$ $(x, y) = (-1, -2)$ is the solution.

Activity 1 : To solve the simultaneous equations by determinant method, fill in the blanks

$y + 2x - 19 = 0$; $2x - 3y + 3 = 0$

Solution : Write the given equations in the form $ax + by = c$

$2x + y = 19$

$2x - 3y = -3$

$$D = \begin{vmatrix} \square & \square \\ 2 & -3 \end{vmatrix} = [\square \times (-3)] - [2 \times (\square)] = \square - (\square)$$

$$= \square - \square = \square$$

$$D_x = \begin{vmatrix} 19 & \square \\ \square & -3 \end{vmatrix} = [19 \times (\square)] - [(\square) \times (\square)] = \square - \square$$

$$= \square$$

$D_y = \begin{vmatrix} \square & 19 \\ 2 & \square \end{vmatrix} = [(\square) \times (\square)] - [(\square) \times (\square)]$

$= \square - \square = \square$

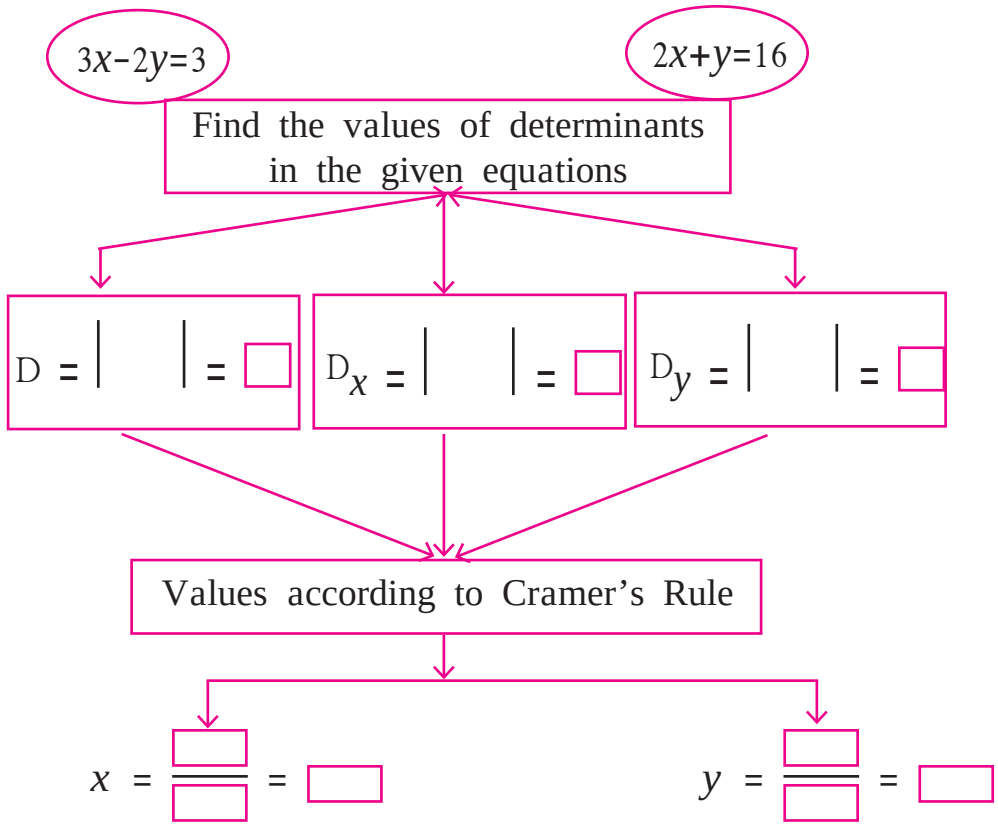
By Cramer's Rule –

$x = \frac{D_x}{D}$ | $y = \frac{D_y}{D}$

$\therefore x = \frac{\square}{\square} = \square$ | $y = \frac{\square}{\square} = \square$

$\therefore (x, y) = (\square, \square)$ is the solution of the given equations.

Activity 2 : Complete the following activity –



$\therefore (x, y) = (\square, \square)$ is the solution.



Let's think.

- What is the nature of solution if D = 0 ?
- What can you say about lines if common solution is not possible?

Practice Set 1.3

1. Fill in the blanks with correct number

$$\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix} = 3 \times \square - \square \times 4 \quad = \square - 8 \quad = \square$$

2. Find the values of following determinants.

(1) $\begin{vmatrix} -1 & 7 \\ 2 & 4 \end{vmatrix}$ (2) $\begin{vmatrix} 5 & 3 \\ -7 & 0 \end{vmatrix}$ (3) $\begin{vmatrix} \frac{7}{3} & \frac{5}{3} \\ \frac{3}{2} & \frac{1}{2} \end{vmatrix}$

3. Solve the following simultaneous equations using Cramer's rule.

(1) $3x - 4y = 10$; $4x + 3y = 5$ (2) $4x + 3y - 4 = 0$; $6x = 8 - 5y$

(3) $x + 2y = -1$; $2x - 3y = 12$ (4) $6x - 4y = -12$; $8x - 3y = -2$

(5) $4m + 6n = 54$; $3m + 2n = 28$ (6) $2x + 3y = 2$; $x - \frac{y}{2} = \frac{1}{2}$



Let's learn.

Equations reducible to a pair of linear equations in two variables

Activity : Complete the following table.

Equation	No. of variables	whether linear or not
$\frac{3}{x} - \frac{4}{y} = 8$	2	Not linear
$\frac{6}{x-1} + \frac{3}{y-2} = 0$		
$\frac{7}{2x+1} + \frac{13}{y+2} = 0$		
$\frac{14}{x+y} + \frac{3}{x-y} = 5$		



Let's think.

In the above table the equations are not linear. Can you convert the equations into linear equations ?



Let's remember!

We can create new variables making a proper change in the given variables. Substituting the new variables in the given non-linear equations, we can convert them in linear equations.

Also remember that the denominator of any fraction of the form $\frac{m}{n}$ cannot be zero.

Solved examples

Solve:

Ex. (1) $\frac{4}{x} + \frac{5}{y} = 7; \frac{3}{x} + \frac{4}{y} = 5$

Solution : $\frac{4}{x} + \frac{5}{y} = 7; \frac{3}{x} + \frac{4}{y} = 5$

$4\left(\frac{1}{x}\right) + 5\left(\frac{1}{y}\right) = 7 \ldots$ (I)

$3\left(\frac{1}{x}\right) + 4\left(\frac{1}{y}\right) = 5 \ldots$ (II)

Replacing $\left(\frac{1}{x}\right)$ by m and $\left(\frac{1}{y}\right)$ by n in equations (I) and (II), we get

$4m + 5n = 7 \ldots$ (III)

$3m + 4n = 5 \ldots$ (IV)

On solving these equations we get

$m = 3, n = -1$

Now, $m = \frac{1}{x} \quad \therefore 3 = \frac{1}{x} \quad \therefore x = \frac{1}{3}$

$n = \frac{1}{y} \quad \therefore -1 = \frac{1}{y} \quad \therefore y = -1$

$\therefore$ Solution of given simultaneous equations is $(x, y) = (\frac{1}{3}, -1)$

Ex.(2) $\frac{4}{x-y} + \frac{1}{x+y} = 3 \; ; \; \frac{2}{x-y} - \frac{3}{x+y} = 5$

Solution : $\frac{4}{x-y} + \frac{1}{x+y} = 3 \; ; \; \frac{2}{x-y} - \frac{3}{x+y} = 5$

$4\left(\frac{1}{x-y}\right) + 1\left(\frac{1}{x+y}\right) = 3 \ldots$ (I)

$2\left(\frac{1}{x-y}\right) - 3\left(\frac{1}{x+y}\right) = 5 \ldots$ (II)

Replacing $\left(\frac{1}{x-y}\right)$ by a and $\left(\frac{1}{x+y}\right)$ by b we get

$4a + b = 3 \ldots$ (III)

$2a - 3b = 5 \ldots$ (IV)

On solving these equations we get, $a = 1 \; b = -1$

But $a = \left(\frac{1}{x-y}\right)$, $b = \left(\frac{1}{x+y}\right)$

$\therefore \left(\frac{1}{x-y}\right) = 1, \left(\frac{1}{x+y}\right) = -1$

$\therefore \; x - y = 1 \ldots$ (V)

$x + y = -1 \ldots$ (VI)

Solving equation (V) and (VI) we get $x = 0, y = -1$

$\therefore$ Solution of the given equations is $(x, y) = (0, -1)$

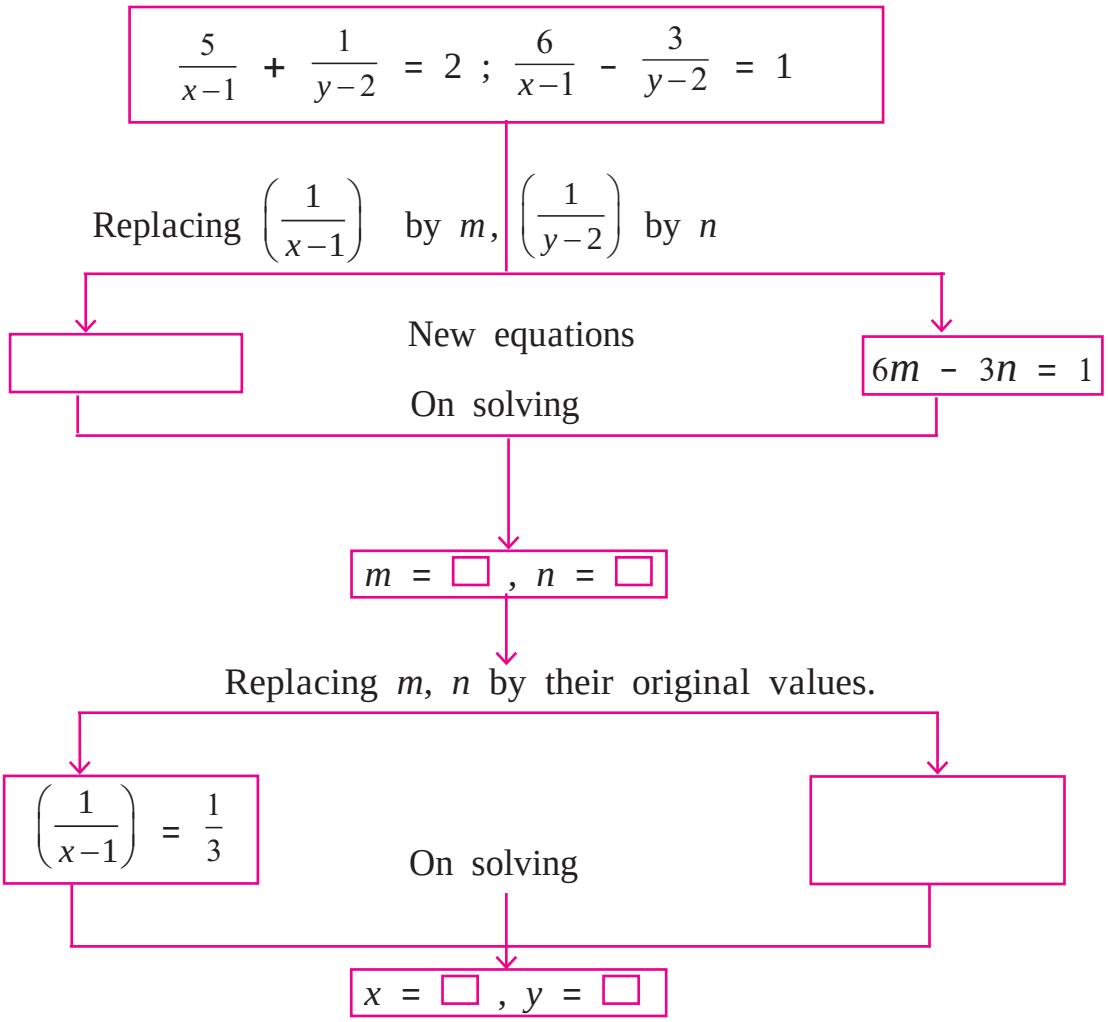


Let's think.

In the above examples the simultaneous equations obtained by transformation are solved by elimination method.

If you solve these equations by graphical method and by Cramer's rule will you get the same answers ? Solve and check it.

Activity : To solve given equations fill the boxes below suitably.



$\therefore$ $(x, y) = (\quad , \quad)$ is the solution of the given simultaneous equations.

Practice Set 1.4

1. Solve the following simultaneous equations.

(1) $\frac{2}{x} - \frac{3}{y} = 15$; $\frac{8}{x} + \frac{5}{y} = 77$

(2) $\frac{10}{x+y} + \frac{2}{x-y} = 4$; $\frac{15}{x+y} - \frac{5}{x-y} = -2$

(3) $\frac{27}{x-2} + \frac{31}{y+3} = 85$; $\frac{31}{x-2} + \frac{27}{y+3} = 89$

(4) $\frac{1}{3x+y} + \frac{1}{3x-y} = \frac{3}{4}$; $\frac{1}{2(3x+y)} - \frac{1}{2(3x-y)} = -\frac{1}{8}$